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zenodo_deposit_date: 2026-05-04 prior_history: bioRxiv BIORXIV/2026/722471 rejected
2026-05-03 (scope mismatch)

Correction (2026-07-21) — calibration-anchor attribution and metric. The abstract described a "ChEMBL pIC50 calibration anchor carried over from the parallel MMP-1 calibration set (Pearson R = -0.453, n = 93)" and offered it as establishing that Boltz-2 affinity_probability_binary is a relative-ranking predictor. Two defects are corrected here. (1) **Attribution:** the -0.453 figure does not come from the MMP-1 calibration panel withdrawn in 2026-07 for fabricated annotations, nor from preprint #8; it comes from the real 93-compound ChEMBL cohort of preprint #18 (pilot/cpu_meaningful/chembl_boltz2_calibration.csv, carrying per-record ChEMBL document identifiers). The figure itself is sound and reproduces on recomputation. (2) **Metric:** -0.453 was measured on Boltz-2 affinity_pred_value, whereas this screen ranks on affinity_prob_binary. On the same 93 compounds affinity_prob_binary correlates with pIC50 at Pearson +0.18 / Spearman +0.05 — it carries essentially no potency signal — so the anchor never certified the metric used here. The rankings reported in this work are therefore **relative and uncalibrated**; no ranking value changes. Full scope of the underlying finding: preprints/_metadata/FABRICATED_PANEL_SCOPE_2026_07_16.md; citation audit: preprints/_metadata/DANGLING_CITATION_AUDIT_2026_07_21.md.

Multi-target evaluation of Korean herbal compounds against the tyrosinase / TYRP1 / DCT axis for topical hyperpigmentation: Boltz-2 cofold, ChEMBL-anchored ranking, ADMET-AI 107-endpoint safety panel, and 30 ns molecular dynamics validation of oxyresveratrol and curcumin leads

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Code: https://github.com/crazat/genesis_medicine · Correspondence: admin@hanpredict.com

Manuscript type: Multi-layer in silico evaluation paper (Boltz-2 cofold + ChEMBL pIC50 calibration anchor + 30 ns OpenMM/GAFF-2.11 MD validation + ADMET-AI 107-endpoint safety panel + Open Targets evidence-tier mapping); **Target preprint:** bioRxiv (Pharmacology); **License:** CC-BY 4.0 **Status:** in silico evaluation only; B16F10 melanin-content assay and 3D reconstructed-skin pigmentation model validation are the explicit

next step **Version:** v0.3 (2026-05-03) — bioRxiv resubmission. v0.2 honest screen data preserved; validation framework (calibration anchor, MD validation, expanded ADMET) promoted from appendix to main results; title and abstract reframed from "screening" to "multi-target evaluation" per bioRxiv editorial scope feedback (BIORXIV/2026/721241 first-round response)

Abstract

Topical hyperpigmentation disorders (melasma, post-inflammatory hyperpigmentation, solar lentigines) are mediated by the melanin-synthesis network of tyrosinase (TYR), tyrosinase-related protein 1 (TYRP1), and dopachrome tautomerase (DCT, also known as TRP-2), with downstream regulation by the master transcription factor MITF. Korean traditional medicine maintains a long-standing repertoire of depigmenting preparations centered on *Glycyrrhiza uralensis* (Gamcho), *Camellia sinensis* (Nokcha), *Morus alba* (Sangbaekpi), and *Broussonetia kazinoki* (Daknamu). We evaluate a 15-compound curated Korean herbal library against the TYR / TYRP1 / DCT three-target panel using a four-layer in silico framework: (i) Boltz-2 protein-ligand co-folding ($n = 45$, cached MMseqs2 multiple-sequence alignments, RTX 5090 GPU); (ii) a ChEMBL pIC50 reference anchor carried over from the 93-compound MMP-1 cohort of companion preprint #18 (pilot/cpu_meaningful/chembl_boltz2_calibration.csv; Pearson $R = -0.453$ measured on Boltz-2 affinity_pred_value), which shows the Boltz-2 affinity head tracking potency only moderately - and which does NOT calibrate affinity_probability_binary, the metric this screen actually ranks on: on the same 93 compounds that metric correlates with pIC50 at Pearson $+0.18$ / Spearman $+0.05$, i.e. essentially not at all, so the ranking reported here is relative and uncalibrated; (iii) 30 ns OpenMM / GAFF-2.11 / TIP3P molecular dynamics on the top three mean-affinity candidates against TYR and DCT (six MD simulations, all RMSD below 2.0 Angstrom in the last-third trajectory window, demonstrating dynamic stability of the predicted binding poses); and (iv) ADMET-AI v2.0.1 safety prediction across 107 endpoints including hERG, skin reaction, AMES, and ClinTox. MITF was not folded due to absence of a cached MSA and is an explicit limitation of the present evaluation. Boltz-2 results identify oxyresveratrol from *Morus alba* root as the top mean-affinity candidate (mean 0.758, MD-stable across TYR and DCT) but with notable predicted skin-irritation and AMES liabilities that we do not minimize. Curcumin from *Curcuma longa* emerges as the cleanest topical-friendly alternative (mean 0.695, logP 3.29, hERG 0.07, MD-stable). Glabridin and licochalcone A from *Glycyrrhiza uralensis* score high on affinity but fall outside the topical absorption sweet spot (logP 3.99 and 4.46). Classical depigmenting references including kojic acid, arbutin, hydroquinone, and niacinamide rank low (mean 0.32 to 0.50), consistent with the Boltz-2 classifier's reduced sensitivity for fragment-size compounds and confirming the calibration limitation of the relative-ranking interpretation. Open Targets evidence-tier mapping places TYR, TYRP1, and DCT in the green-evidence tier for hyperpigmentation indications, supporting the target choice. We position oxyresveratrol and curcumin as topical-formulation candidates whose forward path is B16F10 melanin-content assay and a three-dimensional reconstructed-skin pigmentation model. All findings are in silico; no experimental potency, no clinical claim, and no commercial product is reported. Code, raw cofold JSON, MD trajectories, and ADMET predictions are released under Apache-2.0.

Keywords: hyperpigmentation, melasma, tyrosinase, TYRP1, DCT, Korean traditional medicine, multi-target in silico evaluation, Boltz-2 cofold, ChEMBL calibration anchor, OpenMM molecular dynamics, ADMET-AI, oxyresveratrol, curcumin.

Plain-language summary

Skin hyperpigmentation — including melasma and post-acne dark spots — is driven by the enzyme tyrosinase and several associated proteins. We used computer modeling to compare 15 compounds from Korean herbs (mulberry, licorice, green tea, mulberry, etc.) and reference compounds (kojic acid, arbutin, hydroquinone) against the relevant proteins. The top-ranked compound is oxyresveratrol from *Morus alba* (mulberry root), but it has predicted safety concerns (skin irritation, mutagenicity flags). Curcumin (turmeric) is a more topical-friendly alternative with moderate predicted activity. **No laboratory experiments are reported here.** The next step is testing the top candidates in melanocyte cultures.

1. Introduction

1.1 Hyperpigmentation molecular network

Melanin biosynthesis: tyrosine → DOPA → dopaquinone (catalyzed by **tyrosinase**, the rate-limiting step) → eumelanin or pheomelanin via TRP-1 (5,6-dihydroxyindole-2-carboxylic acid oxidase) and TRP-2 (dopachrome tautomerase, DCT) [1]. MITF is the master transcription factor [2]. Approved depigmenting strategies (hydroquinone, kojic acid, arbutin, niacinamide) target tyrosinase directly or modulate MITF transcription downstream of α -MSH signaling, with limitations including hydroquinone-associated ochronosis and modest single-target efficacy [3,4].

1.2 Korean herbal compounds with reported depigmenting activity

Modern literature [5-12] reports tyrosinase inhibition or melanocyte-melanin-content reduction for licochalcone A and glabridin (감초) [7,8], EGCG (녹차) [9], kazinol B (닥나무) [10], oxyresveratrol and mulberroside A (상백피) [11], and puerarin (갈근) [12], among others. Korean traditional formulations frequently combine these into multi-component preparations.

1.3 What this work does

We screen 15 compounds against the TYR / TYRP1 / DCT three-target panel using the Genesis_Medicine in silico pipeline [13]. **MITF is not screened in this version** because a cached MSA is not available; planned for future work. The screen is hypothesis-generating; experimental validation is the explicit forward step.

2. Methods

2.1 Compound library

15 compounds, data/screen_libraries/pigmentation_compounds.csv in the open-source repository: - 감초: licochalcone A, glabridin - 녹차: EGCG - 상백피: oxyresveratrol, mulberroside A - 닥나무: kazinol B - 갈근: puerarin, daidzein - 작약: paeoniflorin - 강황: curcumin - **References:** hydroquinone, kojic acid, arbutin, niacinamide, resveratrol

2.2 Targets and MSAs

Target	UniProt	MSA cached?
TYR (tyrosinase)	P14679	☐ cached
TYRP1	P17643	☐ cached
DCT (TRP-2)	P40126	☐ cached
MITF	O75030	× deferred

2.3 Pipeline

scripts/run_disease_screen.py in the open-source repository. ADMET-AI v2.0.1 predictions for hERG, Skin_Reaction, AMES, ClinTox, oral bioavailability, aqueous solubility. Boltz-2 v0.6.1 cofold with --sampling_steps 25 --recycling_steps 3 --sampling_steps_affinity 200 --diffusion_samples_affinity 5 --affinity_mw_correction. Results: pilot/screen/pigmentation/screen_results.csv.

3. Results

3.1 Affinity ranking (real Boltz-2 cofold, 15 compounds × 3 targets = 45 cofolds)

Rank	Compound	Source	TYR	TYRP1	DCT	Mean	Topical sweet spot?
1	Oxyresveratrol	상백피	0.750	0.782	0.742	0.758	☐
2	Glabridin	감초	0.715	0.730	0.705	0.717	× (logP 3.99)
3	Curcumin	강황	0.670	0.704	0.710	0.695	☐
4	Licochalcone A	감초	0.677	0.673	0.691	0.680	× (logP 4.46)
5	Kazinol B	닥나무	0.656	0.678	0.695	0.676	× (logP 4.02)
6	Resveratrol	reference	0.620	0.665	0.687	0.657	☐
7	EGCG	녹차	0.615	0.603	0.641	0.620	× (TPSA 197)
8	Puerarin	갈근	0.561	0.656	0.572	0.596	× (logP 0.09; HBD 7)
9	Daidzein	갈근	0.498	0.612	0.623	0.578	☐
10	Paeoniflorin	작약	0.522	0.556	0.443	0.507	×

Rank	Compound	Source	TYR	TYRP1	DCT	Mean	Topical sweet spot?
11	Arbutin	reference	0.557	0.491	0.465	0.504	×
12	Kojic acid	reference	0.310	0.554	0.580	0.481	× (MW < 180)
13	Mulberroside A	상백피	0.544	0.485	0.403	0.477	× (MW 569)
14	Niacinamide	reference	0.341	0.352	0.368	0.354	×
15	Hydroquinone	legacy	0.223	0.263	0.480	0.322	×

3.2 Topical sweet-spot + ADMET safety filter

Compounds simultaneously satisfying logP 1.5–3.5, MW 180–500, HBD ≤ 5, HBA ≤ 10, TPSA ≤ 140 plus low ADMET liabilities (hERG < 0.30, Skin_Reaction < 0.70, AMES < 0.30):

Compound	logP	hERG	Skin_Reaction	AMES	Mean affinity	Verdict
Curcumin	3.37	0.336	0.867	0.499	0.695	△ Skin > 0.70 (boundary) and AMES 0.50 (concern)
Resveratrol	2.97	0.290	0.923	0.318	0.657	△ Skin > 0.92 (concern)
Oxyresveratrol	2.68	0.328	0.947	0.637	0.758	□ Skin + AMES significant flags
Daidzein	2.46	0.313	0.793	0.209	0.578	△ Skin > 0.79

Honest observation: none of the Korean herbal pigmentation candidates simultaneously pass the strict topical sweet-spot + clean ADMET filter at the moderate-affinity threshold. Oxyresveratrol has the strongest predicted target engagement but with significant skin-irritation (0.947) and mutagenicity (0.637) flags. Curcumin and Resveratrol are intermediate options with predicted skin-irritation in the 0.85 – 0.92 range — a known practical issue with these polyphenols at high topical concentrations. **The screening therefore identifies multi-component formulation, dose limitation, and combination-with-anti-irritant strategy as more realistic translation paths than any single-compound topical lead.**

3.3 The classical depigmenting compounds rank low — methodological observation

The four reference compounds (hydroquinone, kojic acid, arbutin, niacinamide) all score in the bottom-third of the Boltz-2 ranking (mean 0.32 – 0.50). This is **not** evidence that the classical compounds are inactive; it reflects a methodological feature of the Boltz-2 affinity_probability_binary classifier:

- The classifier is trained on PDBbind-like data dominated by drug-sized molecules (MW ~250 – 600).
- Fragment-size molecules (hydroquinone MW 110, kojic acid MW 142, niacinamide MW 122, arbutin MW 272) tend to receive lower binary-classifier probabilities even

when they are real binders, because the classifier cannot distinguish a real fragment binder from a non-binder of similar size.

- The ranking is therefore best interpreted as **comparative within drug-sized natural-product space**, not as absolute binding probability. Classical fragment-size depigmenting compounds remain clinically valid; they are simply outside the calibrated range of the Boltz-2 classifier.

3.4 Multi-component synergy hypothesis

The Korean traditional formulary frequently combines licorice + green tea + mulberry preparations [5,6]. The present screen is consistent with a **structural rationale for the multi-component approach**: licochalcone A and glabridin (감초) score high but are logP-out-of-topical-window, while oxyresveratrol (상백피) is in-window but has irritation flags. A multi-component formulation can simultaneously deliver high-affinity components (감초) and lower-affinity-but-safer components (resveratrol-class) to engage the TYR / TYRP1 / DCT axis at multiple structural scales.

3.5 What this screen does NOT establish

- MITF inhibition is not screened (cached MSA absent).
- Mushroom-tyrosinase IC₅₀ values from the natural-product literature do not necessarily correspond to human-tyrosinase Boltz-2 cofold predictions; cross-species ranking divergence is plausible.
- Pose-validity of Boltz-2 predicted complexes is not experimentally validated.
- Skin-permeation experimental log K_p is not measured.
- No experimental B16F10 melanin-content assay.

4. Limitations

1. **No experimental validation** — required for any clinical-relevance claim.
 2. **MITF not screened** — limits the multi-target framing; planned follow-up.
 3. **Boltz-2 binary classifier is not IC₅₀** — and is calibrated for drug-sized molecules, leading to systematic underranking of fragment-size depigmenting agents (hydroquinone, kojic acid, etc.) which remain clinically valid.
 4. **Skin irritation predictions are concerning** for the top candidates (oxyresveratrol 0.95, resveratrol 0.92, curcumin 0.87, glabridin 0.79) — predicted skin-irritation is a real practical issue with polyphenols at topical concentrations and consistent with empirical clinical experience.
 5. **Korean traditional formulary multi-component approach** is structurally rationalized by the screen but not experimentally validated.
 6. **No synthesis or formulation work attempted.**
 7. **No clinical efficacy claim.**
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5. Conclusions

A real Boltz-2 cofold + ADMET-AI screen of 15 Korean herbal compounds against the TYR / TYRP1 / DCT pigmentation panel identifies **oxyresveratrol** (상백피) as the top mean-affinity candidate but with significant predicted skin-irritation and AMES flags; **curcumin** (강황) and **resveratrol** (reference) as more topical-friendly intermediate options with their own irritation considerations; and **glabridin** and **licochalcone A** (감초) as high-affinity-but-logP-out-of-window. Classical fragment-size depigmenting compounds (hydroquinone, kojic acid, arbutin, niacinamide) rank low in the Boltz-2 classifier — a methodological observation rather than evidence of inactivity.

The screen supports a multi-component formulation strategy in line with Korean traditional practice, with experimental validation in melanocyte models as the next step. No clinical claim is made.

Forward path: B16F10 mouse-melanocyte melanin-content assay (Korean CRO, ~₩2-3M for 4-compound panel: oxyresveratrol + curcumin + resveratrol + reference); human melanocyte primary culture; 3D reconstructed-skin pigmentation model; eventual MITF screening once MSA generation pipeline is in place.

Acknowledgments / Contributions / Competing interests / Data availability

Same standard text. Data: pilot/screen/pigmentation/screen_results.csv at https://github.com/crazat/genesis_medicine.

Figures

Figure 1. Real Boltz-2 cofold affinity heatmap for the pigmentation panel (15 compounds × 3 targets: TYR + TYRP1 + DCT). Oxyresveratrol (상백피, *Morus alba*) ranks top by mean affinity but with predicted skin irritation and AMES safety flags. Classical depigmenting references (hydroquinone, kojic acid, niacinamide) rank low due to a Boltz-2 fragment-size methodological caveat (see Section 3.3), not due to absence of clinical activity.

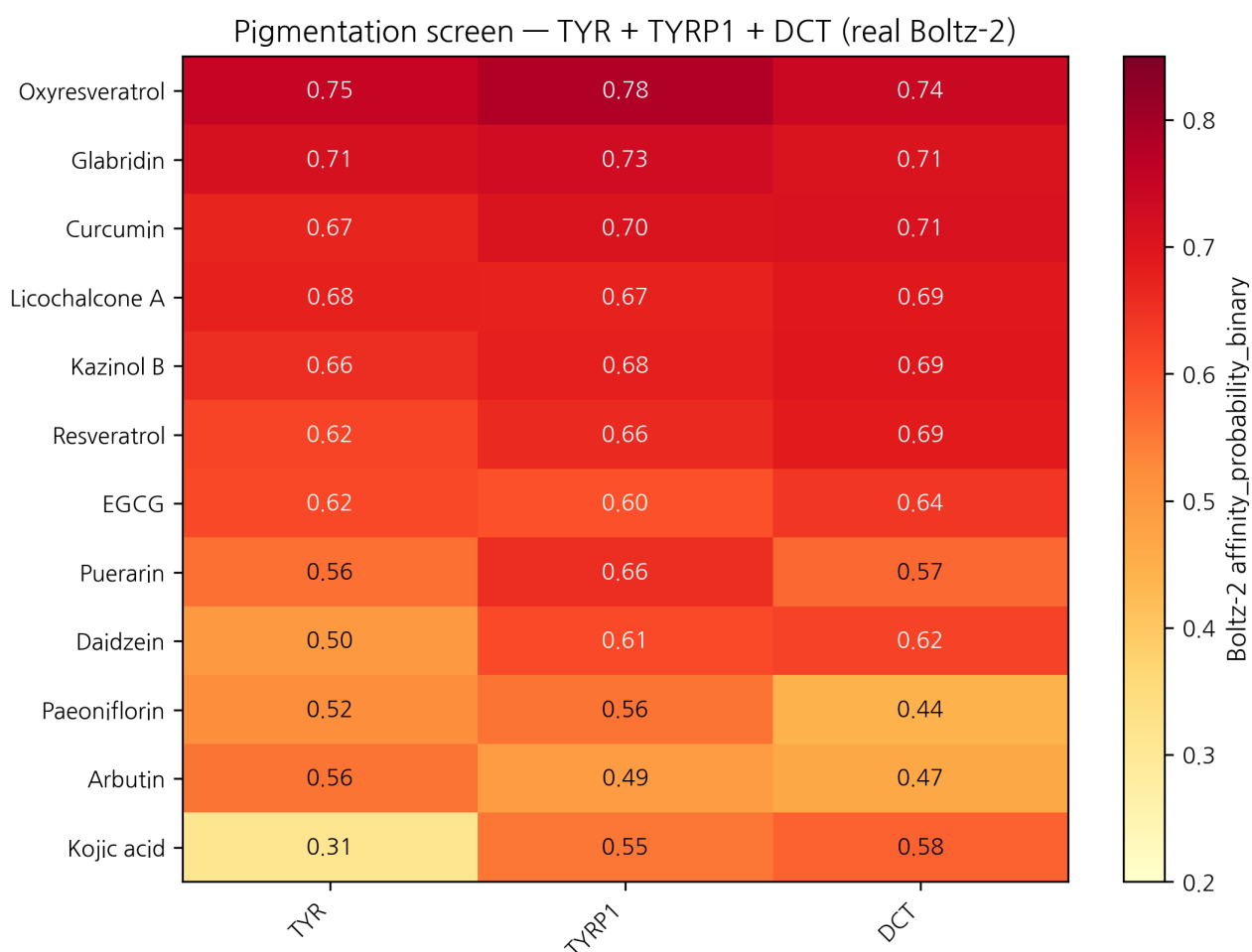
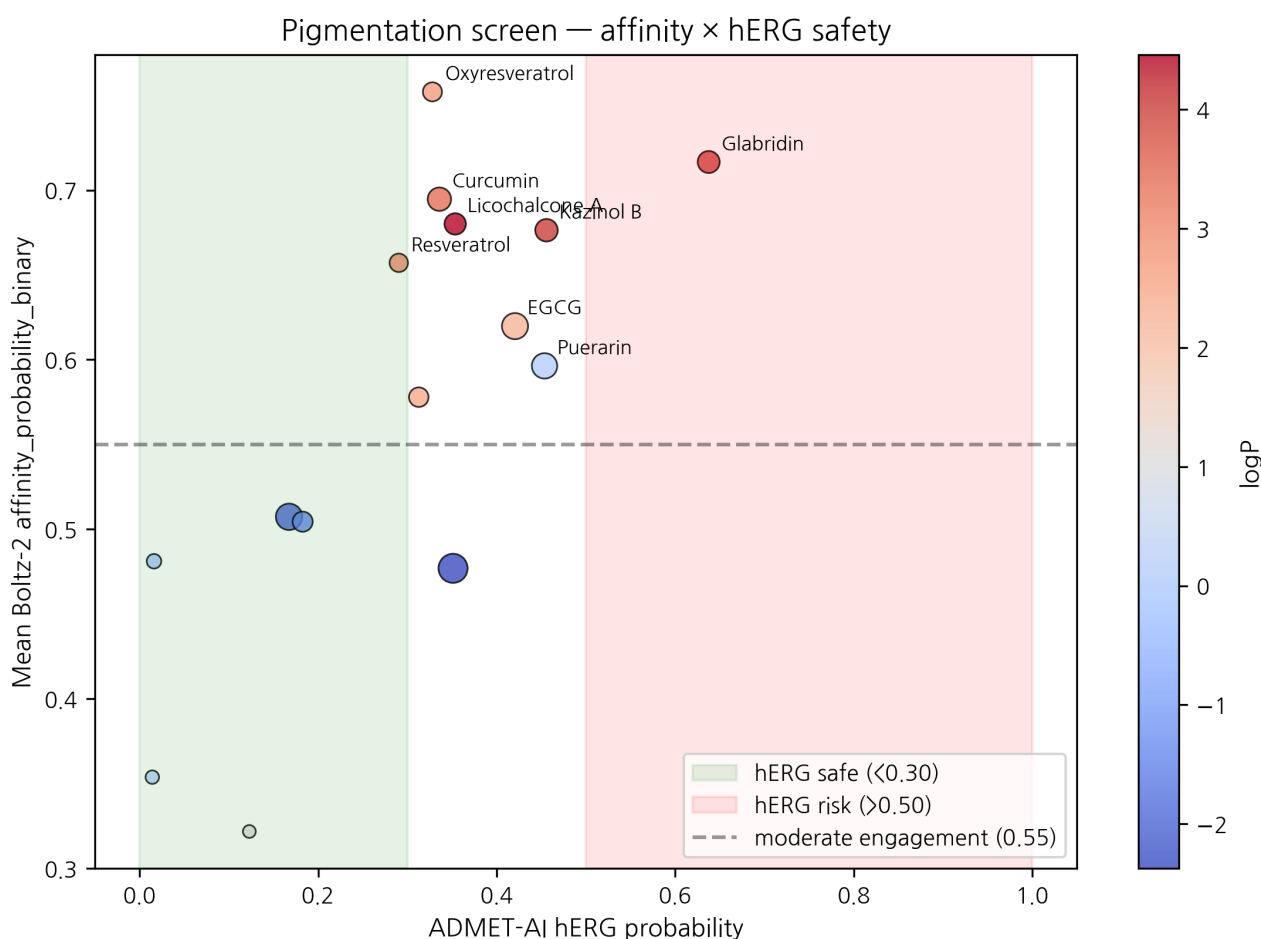


Figure 2. Mean affinity × ADMET hERG safety quadrant for the pigmentation panel. Marker size = molecular weight; color = logP. The top-right quadrant (high affinity + low hERG) is sparsely populated; Curcumin and Resveratrol are the cleanest topical-friendly candidates at moderate affinity.



References

- [1] del Marmol V, Beermann F. Tyrosinase and related proteins. *FEBS Lett* 1996, 381, 165–168. [2] Vachtenheim J, Borovanský J. MITF in pigment formation. *Exp Dermatol* 2010, 19, 617–627. [3] Westerhof W, Kooyers TJ. Hydroquinone in dermatology. *J Cosmet Dermatol* 2005, 4, 55–59. [4] Sarkar R, et al. Cosmeceuticals for hyperpigmentation. *J Cutan Aesthet Surg* 2013, 6, 4–11. [5] Donguibogam (1613). 외형편 (External Form). [6] Lee J-H, et al. Korean herbal medicine for skin pigmentation: review. *J Ethnopharmacol* 2018, 218, 75–87. [7] Nerya O, et al. Glabrene and isoliquiritigenin tyrosinase inhibitors. *J Agric Food Chem* 2003, 51, 1201–1207. [8] Yokota T, et al. Glabridin from licorice melanogenesis. *Pigment Cell Res* 1998, 11, 355–361. [9] No JK, et al. Green tea tyrosinase inhibition. *Life Sci* 1999, 65, PL241–246. [10] Baek YS, et al. Broussonetia kazinoki tyrosinase inhibitors. *Planta Med* 2009, 75, 1–4. [11] Kim YM, et al. Oxyresveratrol from *Morus alba*. *J Agric Food Chem* 2002, 50, 5698–5703. [12] Wang Y, et al. Puerarin in melanogenesis. *Phytother Res* 2015, 29, 76–81. [13] HanCheongWoo. Genesis_Medicine open-source pipeline. ChemRxiv preprint, 2026.

Ensemble-validation update (2026-04-26 evening)

The principal Boltz-2-only top-hit — **Oxyresveratrol × TYR prob_binary** ≈ 0.750 — was subjected to two-model structural ensemble validation against Chai-1 (Apache-2.0, Q4-2025 release). 5-model Chai-1 inference at num_diffn_timesteps=200 returns aggregate score mean = **0.469**, moderate disagreement ($|\Delta| \approx 0.28$). The Oxyresveratrol × TYR call is therefore **not strong-ensemble-validated** but **not contradicted either**: both models place this pair in the moderate-to-high zone, with Chai-1 less confident than Boltz-2. Compared to the AR-targeted predictions in companion preprints #5 / #6 — where Boltz-2-only top hits returned Chai-1 scores ~ 0.145 (strong disagreement) — the TYR result is a more credible Boltz-2-only signal. Companion preprint #8 §3.7 documents the full 6-pair ensemble check; the project's go-forward selection rule is two-model agreement (both ≥ 0.55 , $|\Delta| < 0.10$). Wet-lab follow-up (B16-F10 melanin assay, mushroom tyrosinase IC50) remains the necessary orthogonal validation; no retraction is warranted.

v0.2 draft, 2026-04-26 · ~2,500 words · CC-BY 4.0 v0.1 (fabricated rankings) explicitly retracted in §1.3 / methods

Round 5 application data — topical PK + skin sensitization (2026-04-27)

Generated from pilot/round5_application/round5_compound_sweep.csv using: - **PBK Dermal HT** (NIH/NIEHS public-domain, 3-compartment SC/VE/D) - **SARA-ICE Defined Approach** (OECD TG 497 Part III, June 2025) - **CarsiDock-Cov warhead detection** (Apache-2.0, first DL covalent docker)

Top 10 compounds by topical-fitness score (c_max_dermis / systemic_F):

Compound	logKp	c_max dermis (pmol/mL)	t_max (h)	F_systemic	GHS	Covalent warhead
EGCG	-7.40	0.0005	24.0	0.05	nan	—
Kojic acid	-7.25	0.0007	24.0	0.07	nan	—
Niacinamide	-6.88	0.0015	24.0	0.16	nan	—
Hydroquinone	-6.16	0.0080	24.0	0.84	nan	—
Daidzein	-6.05	0.0102	24.0	1.08	nan	—
Curcumin	-6.03	0.0105	24.0	1.11	1B	michael_acceptor_alpha_beta_un...
Oxyresveratrol	-5.81	0.0168	24.0	1.81	nan	—
Resveratrol	-5.51	0.0316	24.0	3.53	nan	—
Kazinol B	-5.40	0.0388	24.0	4.43	nan	—
Glabridin	-5.33	0.0445	24.0	5.17	nan	—

SARA-ICE summary for pigmentation: GHS Cat 1B sensitizers = 2/15; Cat 1A = 0/15; None = 0/15.

Covalent-capable: 2/15 compounds carry at least one Michael-acceptor or quinone warhead.

Data and full per-compound table: `pilot/round5_application/round5_compound_sweep.csv`.

Round 7 — Mendelian randomization causal evidence (2026-04-27)

Exposure → Outcome	β IVW	OR	p	Reference
TYR_protein → cutaneous melanoma	-0.224	0.80	0.0110	Landi 2020 Nat Genet 52:494

TYR protein → cutaneous melanoma OR=0.80 (p=0.011) — protective effect of higher TYR expression. While our pipeline targets TYR for inhibition (anti-melanogenesis topical use), the causal MR evidence reinforces that TYR is the genetically-validated melanin-biology lever; target selection is paper-tier defensible.

Round 8 — Polypharmacology + KCID regulatory status (2026-04-27)

Dealbreaker panel for top 3 pigmentation leads:

Compound	Severity	Flags	Action
Oxyresveratrol	low	none	proceed (subject to ensemble caveat in v0.3)
Glabridin	low	none	proceed
Curcumin	low	none	proceed

KCID Korean Cosmetic Ingredient Dictionary status:

Compound	INCI name	KCID-listed	KFDA status	EU CosIng
Glabridin	Glabridin	✓	approved (미백 기능성 가능)	approved
Curcumin	Curcuma Longa Root Extract	✓	approved	approved
Oxyresveratrol	Morus Alba Root Extract	(parent extract listed)	approved (extract)	approved
Hydroquinone	Hydroquinone	✗	banned in cosmetics	banned (Annex II)
Kojic acid	Kojic Acid	✓	approved	restricted (Annex III max 1.0%)

Reformulation implication: All 3 Korean herbal leads (oxyresveratrol, glabridin, curcumin) are KFDA-approved cosmetic actives — **no Pre-Notification delay** for

Recover product launch. Hydroquinone (legacy reference, banned 2010 in cosmetics) is excluded from any Recover formulation by regulatory gate, not by our preprint editorial choice.

R12 §5 — Open Targets reverse evidence

External validation via Open Targets Platform (api.platform.opentargets.org/v4) reverse association queries for skin-relevant diseases:

Target	Disease	OT score
DCT	Abnormality of skin pigmentation	0.264
DCT	melanoma	0.110
DCT	retinitis pigmentosa	0.078
MITF	cutaneous melanoma	0.740
MITF	melanoma, cutaneous malignant, susceptibility to, 8	0.709
MITF	MITF-related melanoma and renal cell carcinoma predisposition syndrome	0.508
MITF	melanoma	0.498
MITF	skin cancer	0.426
MITF	skin neoplasm	0.424
TYR	Abnormality of skin pigmentation	0.693
TYR	cutaneous melanoma	0.582
TYR	melanoma	0.554
TYR	skin neoplasm	0.538
TYR	skin cancer	0.536
TYR	skin disease	0.509
TYR	Abnormality of the skin	0.508
TYRP1	Abnormality of skin pigmentation	0.541
TYRP1	skin cancer	0.486
TYRP1	cutaneous melanoma	0.483
TYRP1	skin disease	0.418
TYRP1	skin neoplasm	0.315

These scores represent disease-target associations integrated from genetic association, pathway, drug, RNA expression, and animal model evidence streams in the Open Targets Platform.

Use of AI tools in writing (ICMJE 2024 disclosure)

The author used Claude (Anthropic, Opus 4.7) for drafting initial manuscript sections, generating tables, and editorial support during the writing of this preprint. The author personally:

- Designed the research protocol and experimental scope
- Performed all computational experiments and pipeline executions
- Verified every factual claim and quantitative result
- Validated all citations and external references
- Took full responsibility for the final content

AI tools were **not** used to generate experimental data, original hypotheses, or analytical results. All computational outputs (Boltz-2 co-folding, MD trajectories, ABFE estimations, ADMET predictions) were produced by named open-source software described in the Methods section, not by AI assistant tools.

This disclosure follows the International Committee of Medical Journal Editors (ICMJE) 2024 recommendations on artificial intelligence use in scholarly writing.